

AI Tools
Prompt Engineering
Generative AI



ChatGPT



DeepSeek



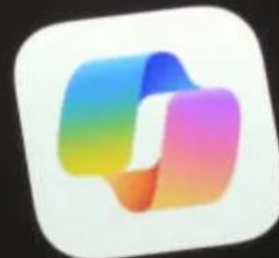
Gemini



Claude



Grok



Copilot

General AI Assistants

- • ChatGPT (OpenAI)
 - Features: Conversational AI, text generation, code, summarization
 - Use Cases: Research, brainstorming, drafting
- Claude (Anthropic)
 - Features: Long context understanding, safe AI responses
 - Use Cases: Document analysis, idea generation
- Gemini (Google DeepMind)
 - Features: Search-integrated AI, multi-modal
 - Use Cases: Research with source citations
- Perplexity AI
 - Features: AI search with citations
 - Use Cases: Fact-checking, quick research

Content Creation & Design

- Canva Magic Studio
 - Features: Text-to-image, Magic Write, presentation generation
 - Use Cases: Social media posts, marketing assets
- Midjourney
 - Features: Text-to-image art generation
 - Use Cases: Graphic design, concept art
- DALL·E 3
 - Features: Realistic and artistic image generation
 - Use Cases: Ad creatives, illustrations
- Stable Diffusion
 - Features: Open-source text-to-image
 - Use Cases: Custom AI image models
- Runway ML
 - Features: Text-to-video, image inpainting
 - Use Cases: AI filmmaking, creative projects
- Pictory
 - Features: Script-to-video
 - Use Cases: YouTube content, explainer videos
- Synthesia
 - Features: AI video avatars
 - Use Cases: Corporate training videos



Coding & Development

- GitHub Copilot
 - Features: AI code suggestions
 - Use Cases: Software development, bug fixes
- Tabnine
 - Features: Predictive coding AI
 - Use Cases: Faster coding in IDEs
- Cursor
 - Features: AI-native IDE
 - Use Cases: Building AI apps directly
- Codeium
 - Features: Free AI code assistant
 - Use Cases: Multi-language code help
- Replit Ghostwriter
 - Features: AI coding inside Replit
 - Use Cases: Web app development

Data & Analytics

- Power BI Copilot
 - Features: AI visuals, DAX suggestions
 - Use Cases: Business dashboards
- Tableau GPT
 - Features: Conversational analytics
 - Use Cases: Data exploration
- DataRobot
 - Features: AutoML platform
 - Use Cases: Predictive modeling
- MonkeyLearn
 - Features: Text classification, sentiment
 - Use Cases: Customer feedback analysis

NLP & Text Processing

- • Hugging Face Transformers
 - Features: Pre-trained NLP models
 - Use Cases: Chatbots, NER, summarization
- OpenAI API
 - Features: Text generation, embeddings
 - Use Cases: AI applications
- QuillBot
 - Features: Paraphrasing, grammar
 - Use Cases: Academic writing
- DeepL Write
 - Features: AI grammar and style improvements
 - Use Cases: Professional editing

Speech & Audio

- • Whisper (OpenAI)
 - Features: High-accuracy transcription
 - Use Cases: Meeting transcripts
- ElevenLabs
 - Features: Realistic AI voices
 - Use Cases: Audiobooks, narration
- Murf AI
 - Features: Voiceovers
 - Use Cases: E-learning videos
- Coqui XTTS
 - Features: Open-source TTS
 - Use Cases: Custom voice models

Image & Video Analysis

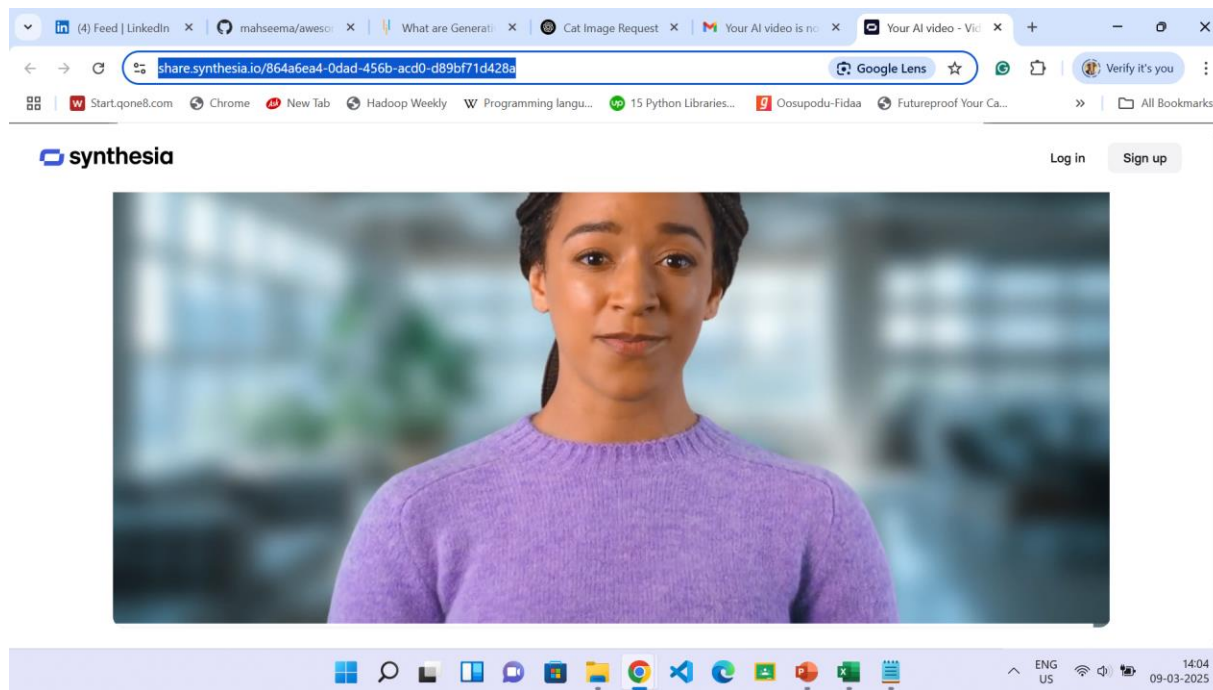
- Google Vision AI
 - Features: Object detection, OCR
 - Use Cases: Document scanning
- Amazon Rekognition
 - Features: Face, scene detection
 - Use Cases: Security, content moderation
- Clarifai
 - Features: Visual AI APIs
 - Use Cases: Image search, tagging
- YOLOv8
 - Features: Real-time object detection
 - Use Cases: Robotics, surveillance

Productivity & Automation

- •Zapier AI Actions
 - Features: Workflow automation
 - Use Cases: Auto-responding to leads
- ClickUp AI
 - Features: Project management
 - Use Cases: Smart task breakdown
- Motion AI
 - Features: AI scheduling
 - Use Cases: Calendar optimization
- Otter.ai
 - Features: Meeting notes & summaries
 - Use Cases: Team collaboration

<https://share.synthesia.io/864a6ea4-0dad-456b-acd0-d89bf71d428a>

<https://share.synthesia.io/864a6ea4-0dad-456b-acd0-d89bf71d428a>





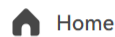
Media AI

AI Video Generator



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AI Effects

Video AI



Image To Video

Veo3



Text To Video

Veo3



Video To Anime

Image AI



Text To Image

VIP Free



Image To Image

VIP Free

Audio AI

Text To Image

Prompt

Sunset by the sea, with golden rays shining



Inspire me

0/1000



Generate

Generated by AI. Please use responsibly.

Sample Video



Prompt:

A silver-hairel



ENG

IN



12:3

11-08-2023

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One of the Greatest Speeches Ever | Steve Jobs Motivation Ark 38M views • 4 years ago

Obama's 2004 DNC keynote speech CNN 2.7M views • 8 years ago

Develop Yourself Day In And Day Out - Les Brown |... LBS 177K views • 3 years ago

Trump vs Zelensky Fight | The End of Ukraine? | Dhruv Rathee Dhruv Rathee 8.3M views • 2 days ago

ENGLISH SPEECH | SUDHA

14:02 09-03-2025

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In this section of Barack Obama's inspirational speech, he shares personal anecdotes to inspire students to prioritize education and overcome challenges. He emphasizes the importance of education in shaping one's future and encourages students to believe in their potential and work hard. Obama highlights the significance of learning from failures and not letting them define one's identity. He advises students to ask for help when needed and never give up on themselves or their country. Obama urges young people to believe in themselves and strive for greatness by making personal contributions and solving problems.

See more



Free Text Summarizer

Modes: Paragraph Bullet Points Summary Length: Short  Long

Enter or paste your text and press "Summarize."


Paste Text



Plagiarism ▾ Proofreading ▾ Citing ▾ AI Writing ▾ Learn

Modes: Paragraph Bullet Points Summary Length: Short Long

Tariffs have historically served a key role in the trade policy of the United States. Their purpose was to generate **revenue for the federal government and** to allow for **import substitution industrialization** (industrialization of a nation by replacing imports with domestic production) by acting as a protective barrier around infant industries. [1] They also aimed to reduce the trade deficit and the pressure of foreign competition. **Tariffs were one of the pillars of the American System** that allowed the **rapid development and industrialization** of the United States.[2]

The United States **pursued a protectionist policy from the beginning of the 19th century until the middle of the 20th century**. Between 1861 and 1933, they had one of the highest average tariff rates on

Select keywords ⓘ

United States

federal government

UK

federal

tariffs

Tariffs have played a crucial role in the US trade policy, generating revenue for the federal government and facilitating import substitution industrialization. They aimed to reduce the trade deficit and foreign competition pressure. Tariffs were a pillar of the American System, enabling rapid development and industrialization. The US pursued a protectionist policy from the 19th century until the mid-20th century, with one of the highest average tariff rates on manufactured imports. Dartmouth economist Douglas Irwin explains that tariffs have served three primary purposes: raising revenue for the government, restricting imports, protecting domestic producers from foreign competition, and reaching reciprocity agreements. From 1790 to 1860, average tariffs increased from 20% to 60% before declining again to 20%. From 1861 to 1933, tariffs increased to 50%, and from 1934 onwards, they declined substantially to 5%. Since 1935, tariff income has continued to be a declining percentage of

- <https://www.scribbr.com/ai-humanizer/>

Ai text in to human like text

The screenshot shows the Scribbr AI Humanizer web application in a browser window. The browser's address bar shows the URL `scribbr.com/ai-humanizer/`. The page features the Scribbr logo and a navigation menu with links for Plagiarism, Proofreading, Citing, AI Writing, and Learn. The main heading is "AI Humanizer", followed by a description: "Scribbr's AI Humanizer turns AI text into human-like writing by removing common AI patterns." Below this, there are language selection tabs (English (US), French, Spanish, German, All) and a "Modes" section with "Standard" selected, "Fluency" as an option, and a "Synonyms" slider. The interface is split into two columns. The left column contains the input text: "AI (Artificial Intelligence) refers to the development of computer systems that are capable of performing tasks that would typically require human intelligence. These tasks can range from simple ones like recognizing speech or images to more complex ones like decision-making, problem-solving, and even creative endeavors like writing or generating artwork. AI is broadly categorized into two types:". The right column shows the output text, which has been rephrased to sound more natural, with some words highlighted in orange and blue. The output text is: "The creation of computer systems that can carry out operations that ordinarily call for human intelligence is known as artificial intelligence, or AI. These tasks can be as basic as identifying speech or images, or they can be as complex as making decisions, solving problems, or even producing art or writing. AI can be broadly divided into two categories:". The browser's taskbar at the bottom shows various application icons and the system clock indicating 14:12 on 09-03-2025.

AI Humanizer

Scribbr's AI Humanizer turns AI text into human-like writing by removing common AI patterns.

English (US) French Spanish German All

Modes: **Standard** Fluency Synonyms:

AI (Artificial Intelligence) refers to the development of computer systems that are capable of performing tasks that would typically require human intelligence. These tasks can range from simple ones like recognizing speech or images to more complex ones like decision-making, problem-solving, and even creative endeavors like writing or generating artwork. AI is broadly categorized into two types:

The creation of computer systems that can carry out operations that ordinarily call for human intelligence is known as artificial intelligence, or AI. These tasks can be as basic as identifying speech or images, or they can be as complex as making decisions, solving problems, or even producing art or writing. AI can be broadly divided into two categories:

text2sql

- <https://app2.text2sql.ai/>

Assistant

New Chat

Templates

History

Configuration

Databases

API Settings

Settings

Help

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R krsprojects99@gmail.com

total sales form products table group by region

To calculate the total sales from a **products** table grouped by region, you would need a table that includes both sales and region information. Assuming the **products** table has columns **region** and **sales**, the SQL query would look like this:

```
SELECT
  region,
  SUM(sales) AS total_sales
FROM
  products
```

Copy

Enter your follow-up here...

Send

Windows taskbar: Start, Search, Task View, File Explorer, Google Chrome, VS Code, Edge, Teams, PowerPoint, Excel, Outlook

System tray: ENG IN, 15:25, 09-03-2025

- TutorAi

Introduction to Python Programming Generic

This course provides an in-depth introduction to Python programming. Designed for beginners, it covers fundamental concepts and techniques, leading to more advanced topics. By the end of the course, learners will have a solid understanding of Python and be able to write, debug, and execute Python programs.

1. Getting Started With Python

In this module, learners will be introduced to Python, its history, and its applications. They will learn how to install Python and set up their development environment. Key learning outcomes include understanding the basic syntax and structure of Python programs.

🕒 1h 30m 📖 5 Lessons

2. Python Basics

This module covers the fundamental concepts of Python programming, including variables, data types, and operators. Learners will write simple programs and develop a strong foundation in handling basic data structures.

🕒 1h 30m 📖 5 Lessons

3. Control Flow In Python

Learners will explore control flow statements such as if-else conditions, loops, and list comprehensions. This module will enable learners to write more complex programs with decision-making capabilities.

🕒 1h 30m 📖 5 Lessons

Tutor AI

Current Course

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Module 1: Getting Started with Python

1.1 Introduction to Python



Content

Simplify

Quiz

Example

1.1.1 Introduction

Python is a versatile and powerful programming language that has gained immense popularity since its inception. In this lesson we will explore the history of Python its creator Guido van Rossum and the philosophy behind the language. We will also discuss why Python is widely used and its key applications in various fields. By the end of this lesson you will have a solid understanding of Python's background and its significance in the programming world.

Python's simplicity and readability make it an excellent choice for beginners and experienced programmers alike. Its extensive libraries and frameworks support a wide range of applications from web development to data science and artificial intelligence.

< Previous Lesson

Mark as Complete

Next Lesson >

Introduction to Python Programming Generic

- ▼ 1. Getting Started with Python
 - 1.1 Introduction to Python ✓
 - 1.2 Installing Python
 - 1.3 Setting Up Your Development Environment
 - 1.4 Python Syntax and Structure
 - 1.5 Writing Your First Python Program
 - 1.6 Basic Input and Output
 - 1.7 Overview of Python Data Types
 - 1.8 Introduction to Variables
- 2. Python Basics
- 3. Control Flow in Python
- 4. Functions and Modules
- 5. Data Structures



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[Themes](#)
[Playlists](#)
[Instruments](#)
[BPM](#)

BETA

Generate track Search by reference beta

1. Enter prompt or upload image

Type anything in English



2. Set type

Track

3. Set duration

00 : 45

from 5 sec to 25 min

or choose:

[Genres](#)

[Moods](#)

[Activities](#)

Generate track

Generated tracks

Downloaded





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Moods

Genres

Themes

Playlists

Instruments

BPM

BETA

Generate track

Generated tracks

Downloaded ☐



Gaali Lo Oogisalaade Dhee...

Text-To-Music

110

D

Track

0:45



Gaali Lo Oogisalaade Dhee...

Text-To-Music

110

C

Track

0:45



Home / My Generated Tracks



ENG
IN



16:06
09-03-2025

- Dulingo language Tool
- Writing Tools

mathematical calculations and computations

- <https://www.wolframalpha.com/>
- <https://www.symbolab.com/>

Effective prompt/with upfront

- **Provide Information Upfront:** Including necessary and helpful information at the beginning of your prompt can significantly improve the output.
- Without upfront information
- **Prompt:** Write a witty introduction as a speaker about me.
- **AI's Response:** The AI gives a general introduction about itself
- because it lacks personal information about the user.
- With upfront information
- **Prompt:** [Inserts resume information] Write a witty introduction as a speaker about me.

Effective prompt/with upfront

- Summarizing an Article
- Without upfront information
- **Prompt:** Summarize the article on Cricket Analytics.
- **AI's Response:** The AI won't be able to generate a summary because it doesn't know the content of the specific article you're referring to.
- With upfront information
- **Prompt:** [Inserts article content] Summarize the above article.

Effective prompt/with examples

- Write a Python program to calculate the factorial of a given number.
- Write a Python program to calculate the factorial of a given number.
For example, if the input is 5,
- the output should be 120.

Effective prompt/response length

- Explain the concept of climate change.
- Provide an essay-length explanation (800-1000 words) of the concept of climate change.
- Create a social media post about our new product [name].
- Craft a social media post (maximum 280characters) to promote our new product[name]
- Summarize the key findings of the research paper on artificial intelligence.
- Summarize the key findings of the research paper on artificial intelligence in four to six
- sentences.

Effective prompt/Be Concise and Avoid Multiple Questions

- Avoid asking multiple questions in a single prompt; it can lead to confusion and inaccurate responses.
- Keep your questions concise and focused on one topic to improve clarity and accuracy.

Effective prompt/Using Action Verbs in Prompts

Action Verbs		
Analyze	Explain	Brainstorm
Compare	Summarize	Recommend
Create	Evaluate	Design
Solve	Predict	Demonstrate
Illustrate	Investigate	Describe
Interpret	Critique	Develop
Implement	Justify	Outline
Investigate	Define	Propose
Formulate	Identify	Assess

Effective prompt

- **Analyzing:** "Analyze the factors contributing to climate change and its impact on global ecosystems."
- **Explaining:** "Explain the process of photosynthesis in plants and its significance in the ecosystem."
- **Brainstorming:** "Brainstorm innovative ideas for reducing plastic waste in urban areas."
- **Comparing:** "Compare and contrast the advantages and disadvantages of renewable energy sources."
- **Summarizing:** "Summarize the key findings of the research paper on artificial intelligence advancements."
- **Providing Recommendations:** "Provide recommendations for improving employee engagement in a remote work setting."

Effective prompt/Tone of Voice in Prompts

- Maintaining a consistent tone of voice in your prompts helps convey your brand identity effectively.
- **Aggressive:** "Write a blog on the effect of poverty on the economy in an aggressivetone."
- **Calm:** "Compose a blog post discussing the impact of mindfulness on mental health in a calm and composed tone."
- **Confident:** "Craft an article about personal finance strategies with a confident and authoritative tone."
- **Excited:** "Write a review of the latest science fiction novel, capturing your excitement and enthusiasm for its imaginative plot."
- **Formal:** "Develop a research paper on the historical significance of the Industrial Revolution using a formal and scholarly tone."

Effective prompt/"Pretend You Are an Expert"

- What exercises should I do to improve my core strength?
- Pretend you are a fitness expert. What exercises should I do to improve my core strength?
- I'm planning a budget-friendly trip to Paris. Any tips?
- Pretend you are a travel expert. I'm planning a budget-friendly trip to Paris. Any tips?

Effective prompt/Play Devil's Advocate

- Use this phrase to get unique and specific answers.
- Forces ChatGPT to consider counterarguments.
- Deeper analysis of problem statements.
- Useful for making more informed decisions

Effective prompt/ "Pretend You Are My Consultant"

- Treat ChatGPT as a personal consultant.
- Provides detailed insights and tailored advice.
- Similar to consulting from experts.
- Helps in making better decisions.

Back bone of Generative AI

- Neural Networks
- Convolutional Neural Network
- Recurrent Neural Network
- Generative Adversarial Networks
- Transformers
- NLP

Diffusion / Generative Adversarial Networks

- Diffusion models gradually transform data from simple into a complex one and generate realistic art and images and content creation
- GAN consists of two neural networks
 - Generator
 - Discriminator

- **Diffusion Models** → Best for **image/video** generation.
- **Seq2Seq Models** → Great for **translation & summarization**.
- **BERT-based Models** → Best for **text understanding** (search, classification).
- **GPT-based Models** → Best for **text generation & chatbots**.

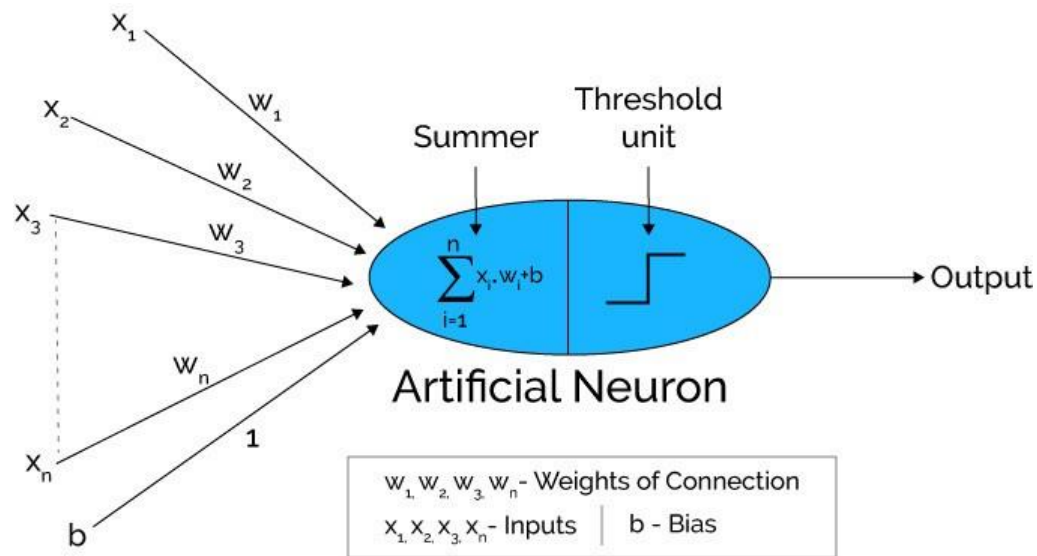
Model Type	Use Case	Works With	Directionality	Best For
Diffusion	Image & video generation	Noise to structure	Iterative refinement	Art, deepfake generation
Seq2Seq	Translation, summarization	Input → Output	Encoder-Decoder	Language translation, chatbots
BERT	Text understanding	Context-based	Bidirectional	Search engines, NLP understanding
GPT	Text generation	Predicts next word	Unidirectional	Chatbots, content generation

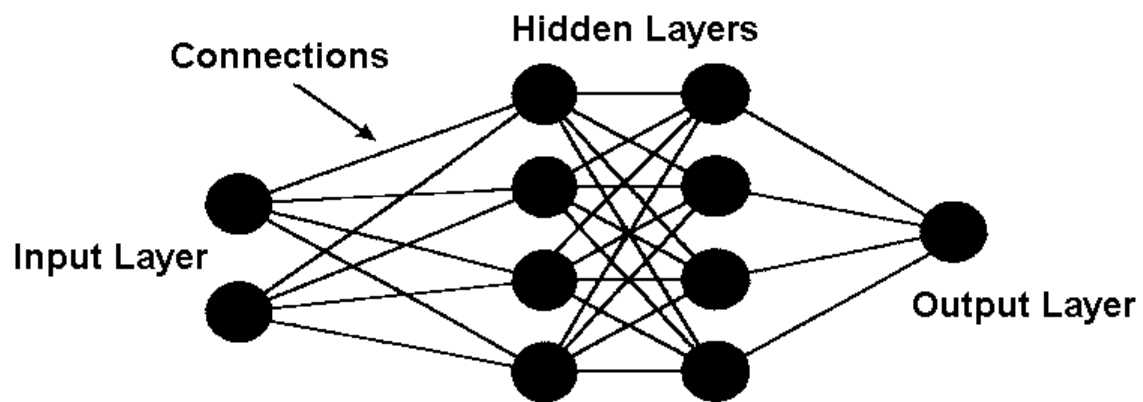
Diffusion Models	Stable Diffusion	Stability AI	Text-to-Image	Latent Diffusion Model (LDM)	AI Art, Image Generation
	DALL-E 2	OpenAI	Text-to-Image	Diffusion with CLIP Guidance	Creative AI, Concept Art
	Imagen	Google	Text-to-Image	High-quality diffusion	Photorealistic Image Generation
	Sora	OpenAI	Text-to-Video	Advanced latent diffusion	AI-powered video generation
	LDM (Latent Diffusion Model)	LMU Munich	Image ↓ thesis	Efficient diffusion models	AI Art, Style Transfer

Seq2Seq Models	T5 (Text-to-Text Transfer Transformer)	Google	Text Processing	Encoder-Decoder	Summarization, Translation
	BART (Bidirectional Auto-Regressive Transformer)	Facebook (Meta)	Text Completion	Denoising Autoencoder	Summarization, Text Generation
	mBART (Multilingual BART)	Facebook (Meta)	Translation	Pretrained on multiple languages	Machine Translation
	Marian NMT	Microsoft	Translation	Transformer-based NMT	Multilingual AI

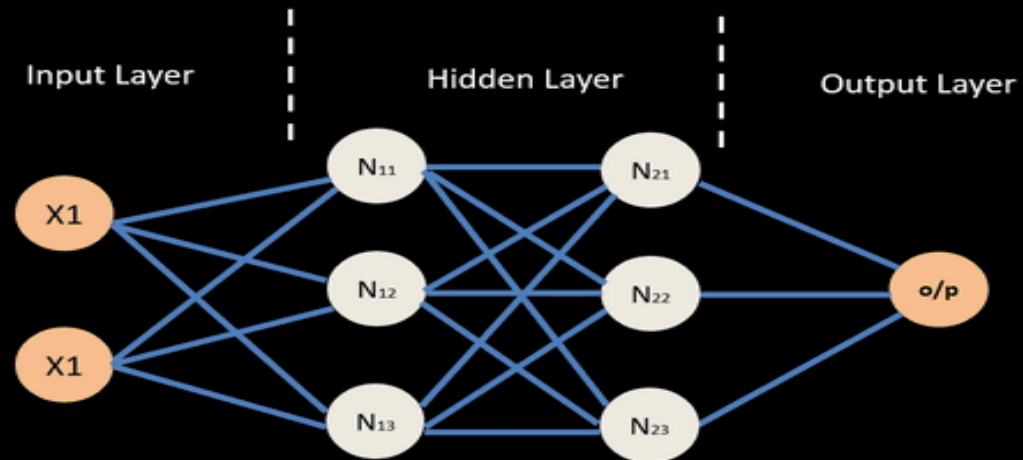
BERT-based Models	BERT (Bidirectional Encoder Representations from Transformers)	Google	Text Understanding	Context-aware bidirectional learning	Search, NLP Understanding
	RoBERTa	Facebook (Meta)	Text Analysis	Optimized BERT variant	Sentiment Analysis, Text Processing
	ALBERT (A Lite BERT)	Google	NLP Efficiency	Compressed BERT	NLP tasks, QA Systems
	DistilBERT	Hugging Face	Lightweight NLP	Faster & smaller BERT	Chatbots, Mobile NLP
	XLNet	Google & CMU	↓ Understanding	Autoregressive pretraining	Question Answering, Text

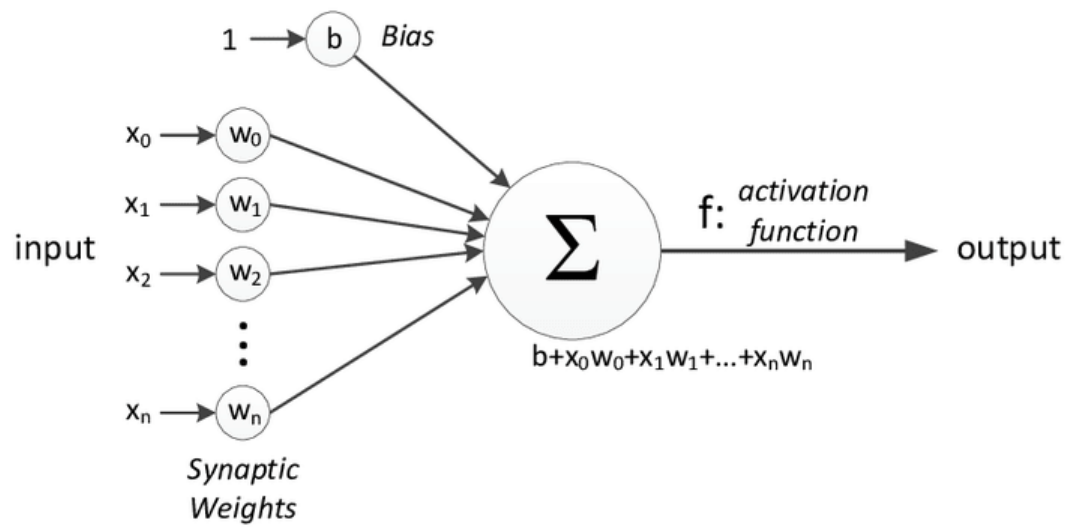
GPT-based Models	GPT-3	OpenAI	Text Generation	Large-scale pretraining	Content Creation, Chatbots
	GPT-3.5	OpenAI	Chat & NLP	Fine-tuned version of GPT-3	ChatGPT, AI Assistants
	GPT-4	OpenAI	Advanced Text AI	Improved reasoning & multimodal	AI Research, Coding
	GPT-4 Turbo	OpenAI	Faster & cheaper	Optimized GPT-4	Chatbots, AI Assistants
	Codex	OpenAI	↓ Generation	GPT-3 fine-tuned for code	GitHub Copilot, AI Coding





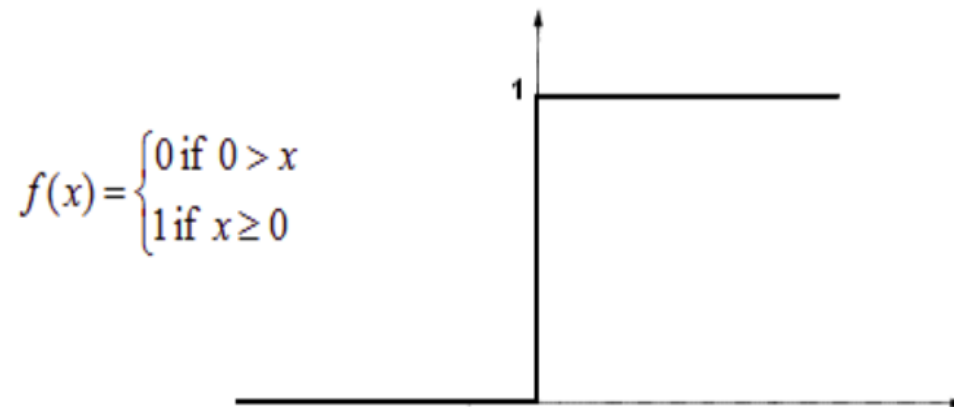
Neural Network – Backpropagation





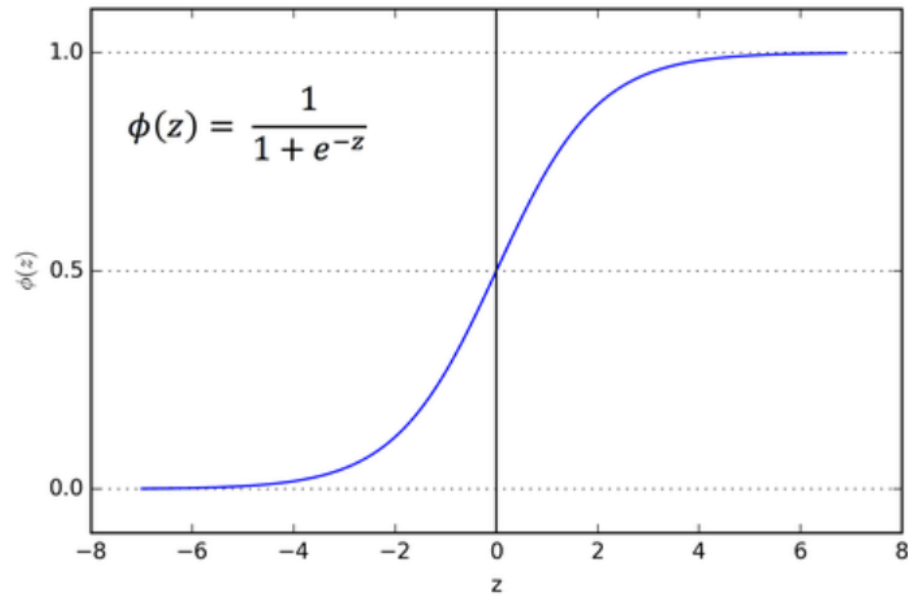
1. Threshold Activation Function

In a threshold activation function, if the input value is above or below a particular threshold, then the neuron is activated and it sends the same signal to the next layer.



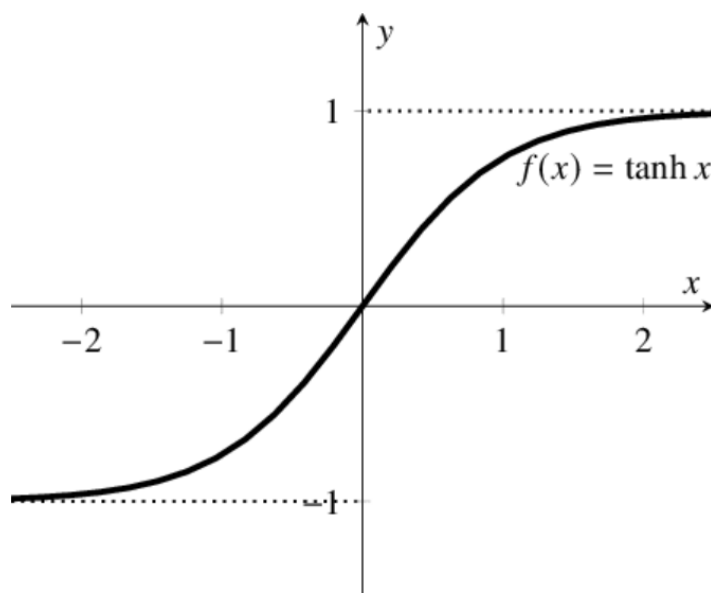
2. Sigmoid Activation Function - (Logistic function)

A sigmoid function gives values between 0 and 1. Hence, it is used in models wherein we need to predict the probability as an output.



3. Hyperbolic Tangential Function - ($\tanh$)

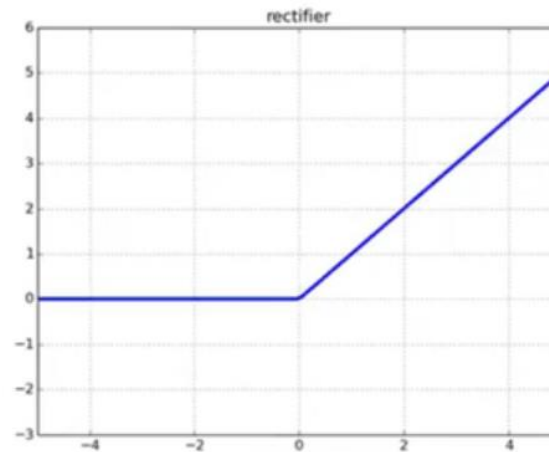
The $\tanh$ function is similar to the sigmoid function but it is better in performance. The range of this function is between $(-1, 1)$.



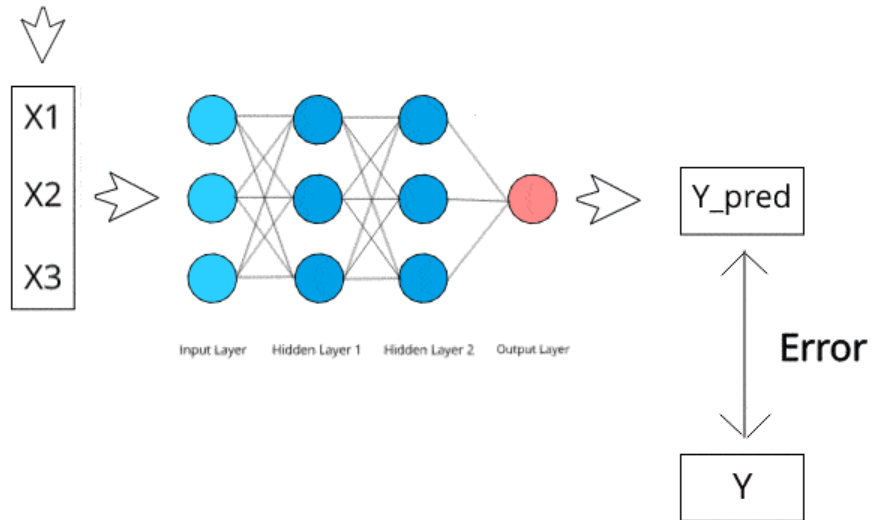
4. Rectified Linear Units - (ReLU)>

ReLU is one of the most widely used activation functions while training a CNN or ANN. The value ReLU ranges from zero to infinity.

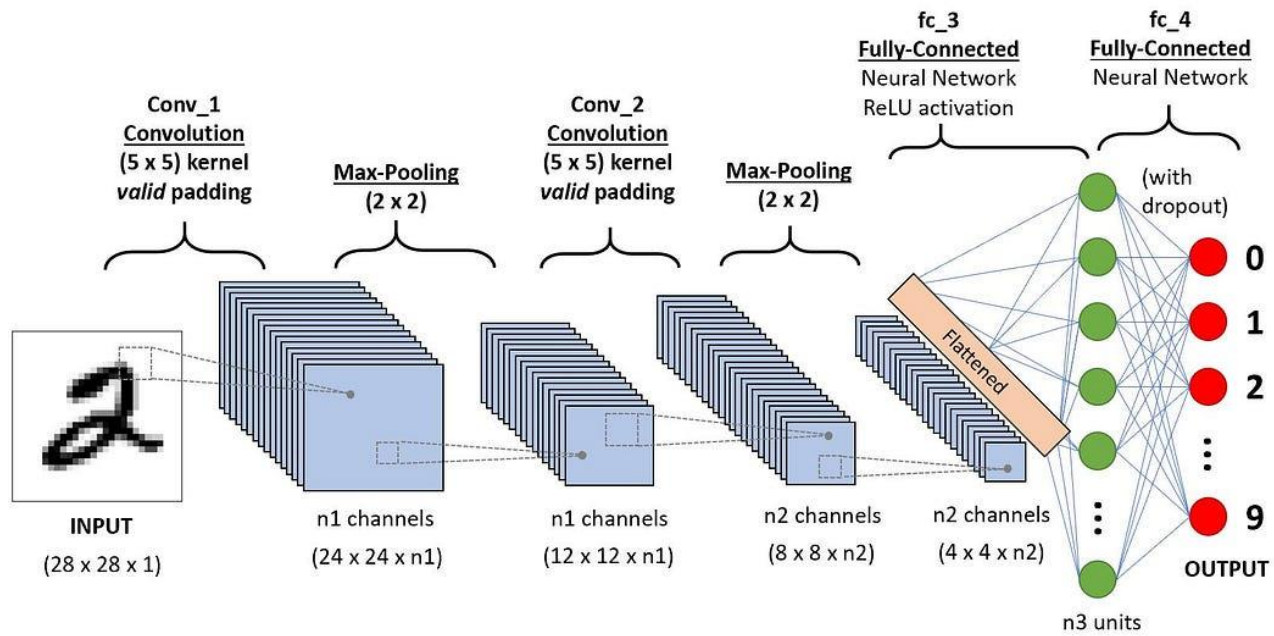
$$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$$



Feed new data



cnn



4x4x3

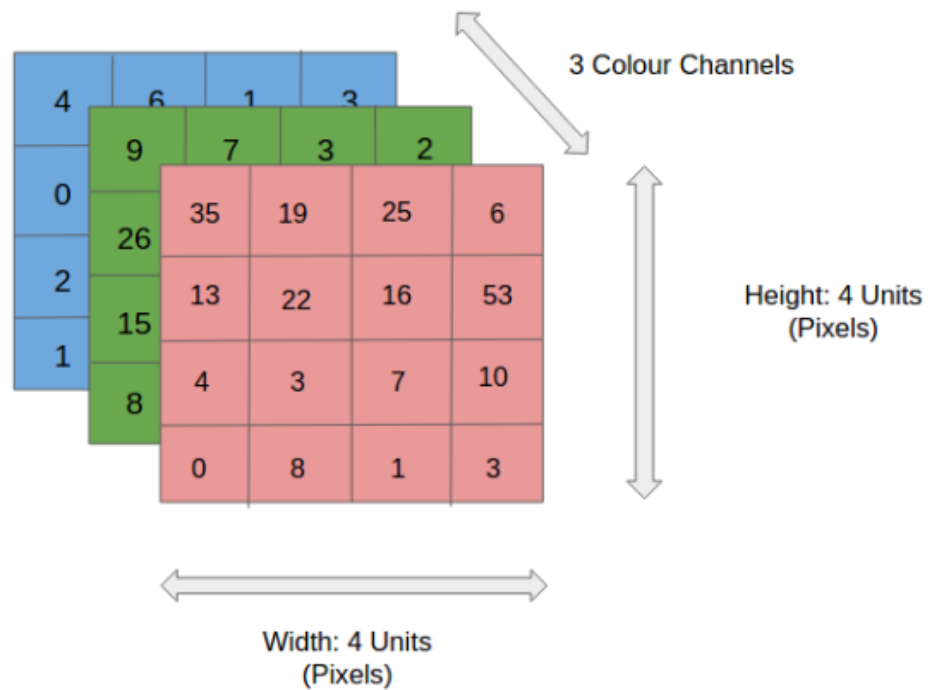
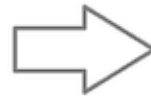


Image to vector

1	1	0
4	2	1
0	2	1



1
1
0
4
2
1
0
2
1

5x5x1 to 3x3x1

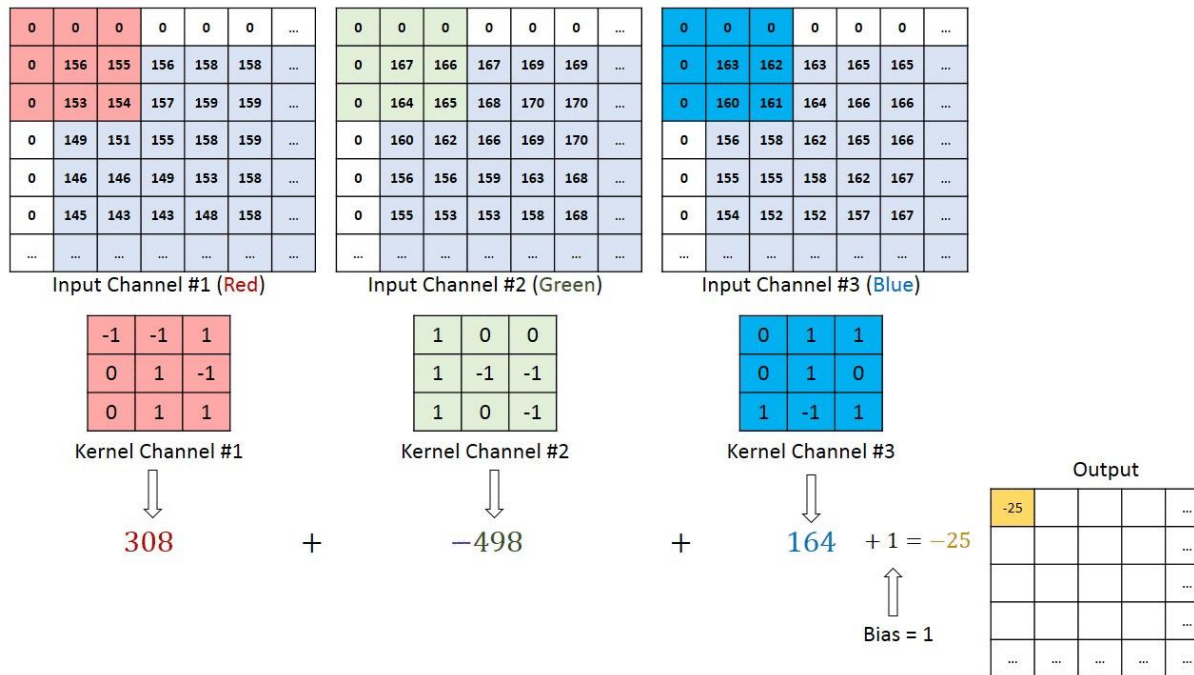
1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0

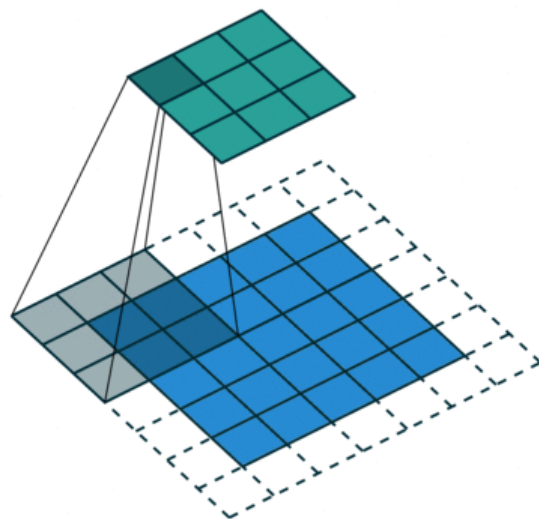
Image

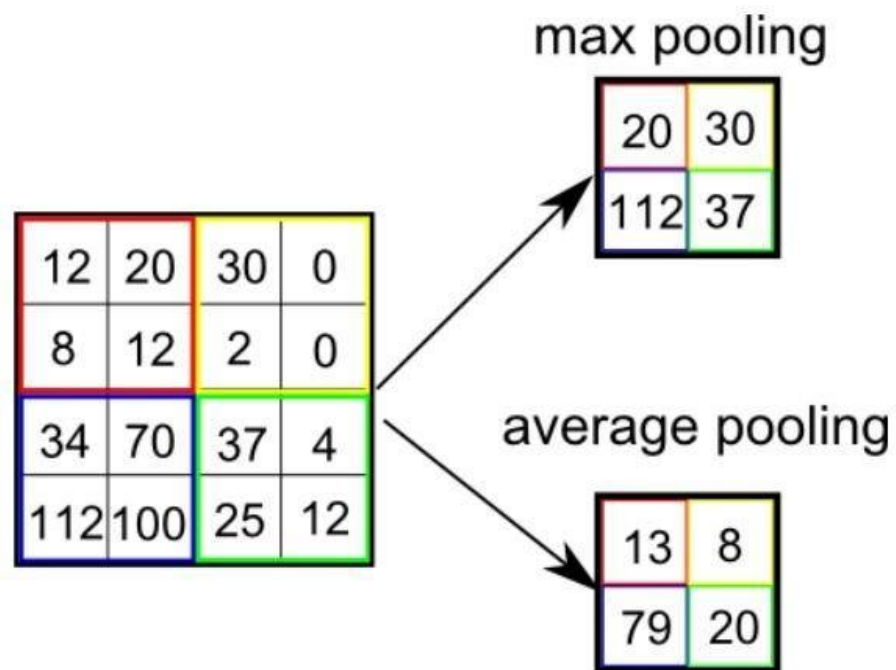
4		

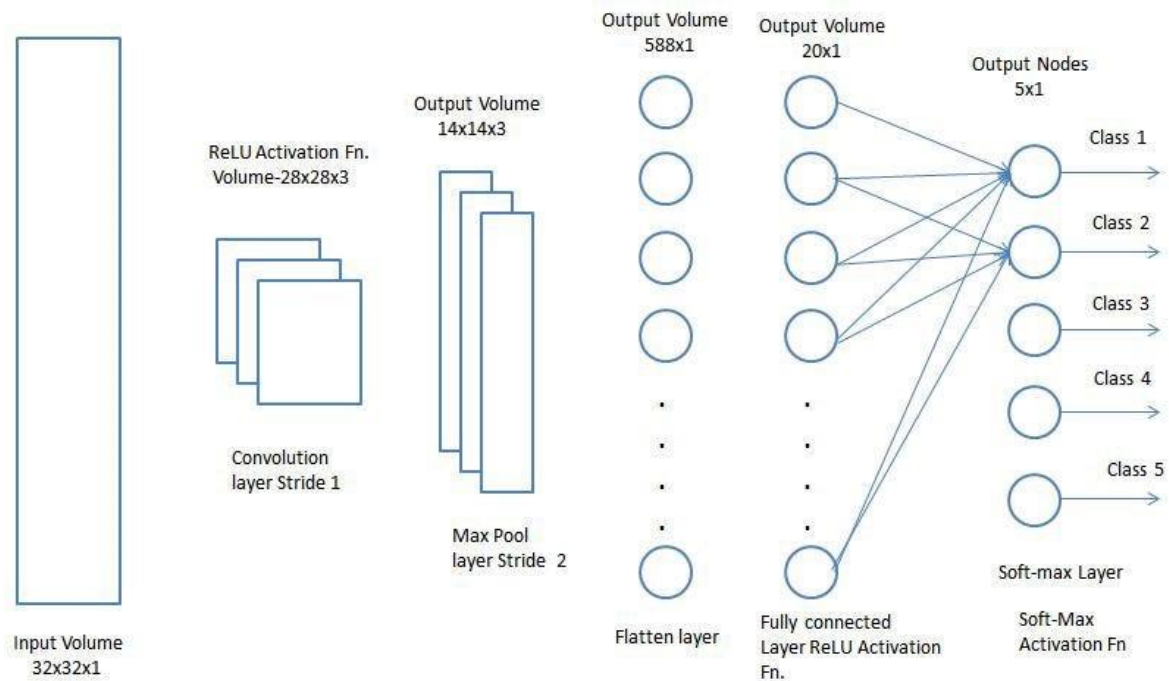
Convolved
Feature

Mxnx3 to 3x3x3

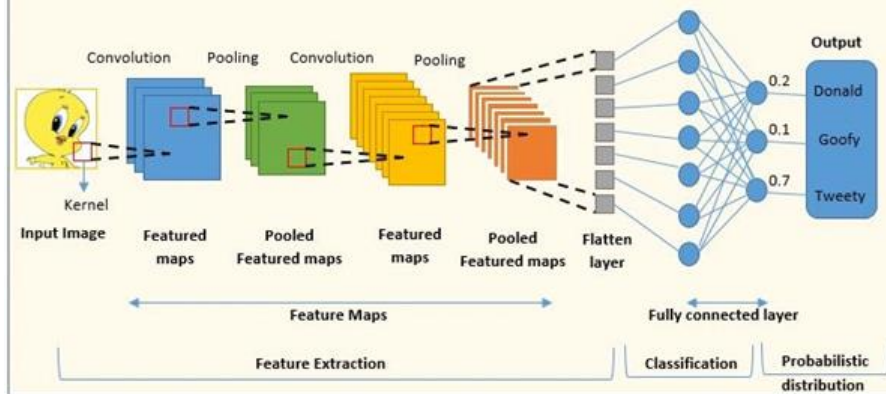


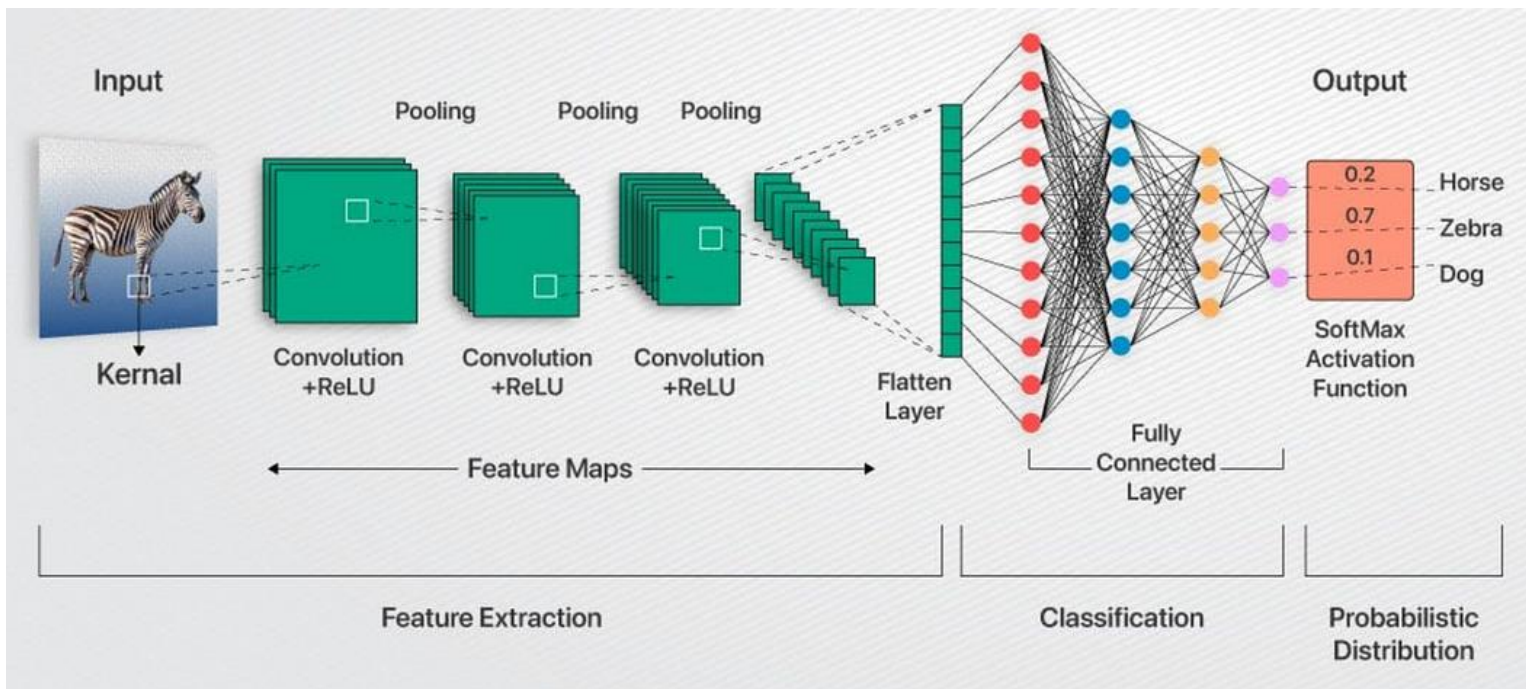




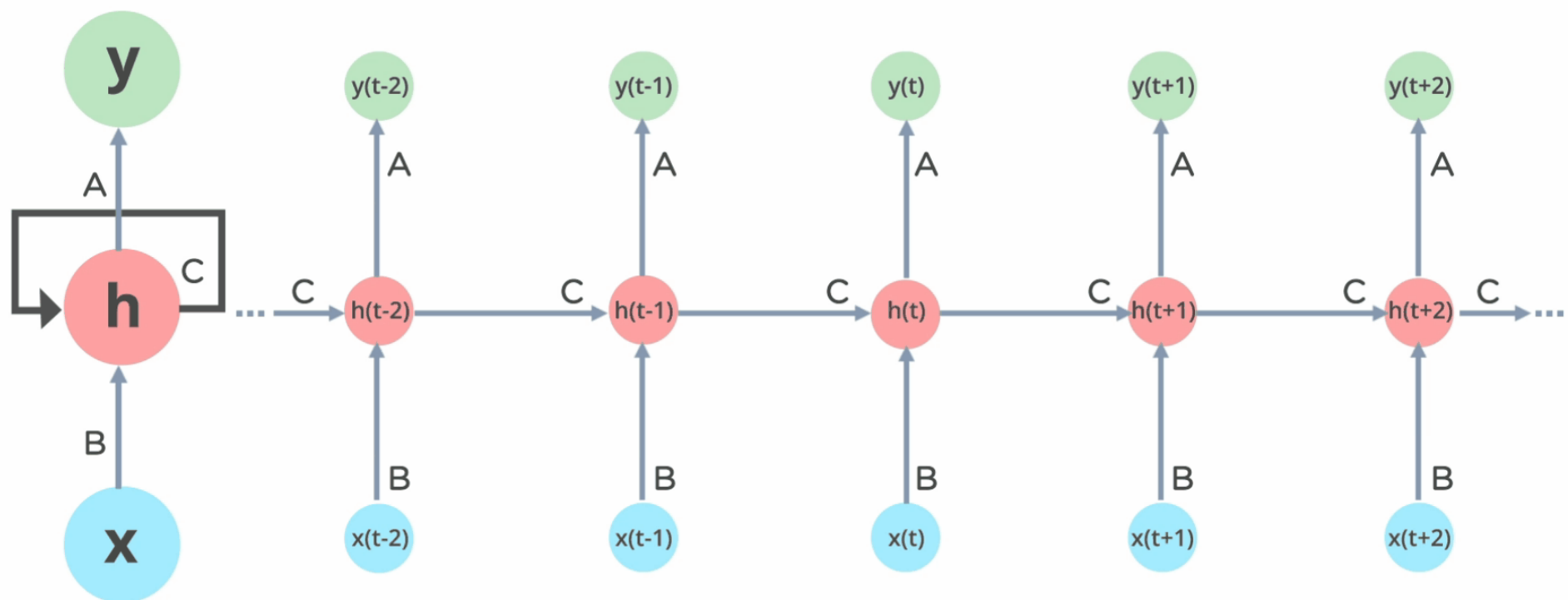


A Typical Convolutional Neural Network (CNN)





rnn



Recurrent Neural Network (RNN)

Input Sequence

X

3	4	5	6
---	---	---	---

Parameters

A

1	-1
1	1

 B

1
2

 C

-1	1
----	---

Activation Function

ϕ : ReLU

Hidden States

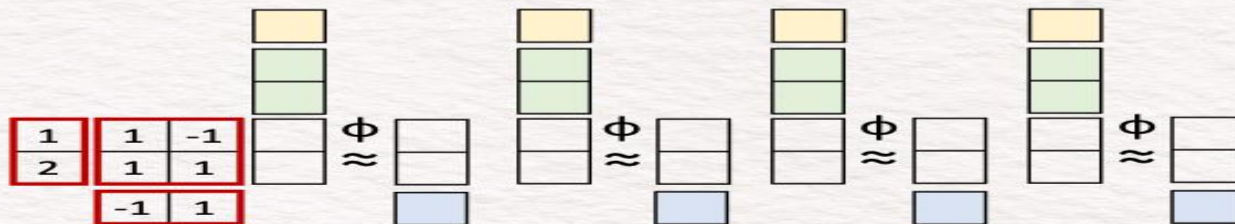
H_0

0
0

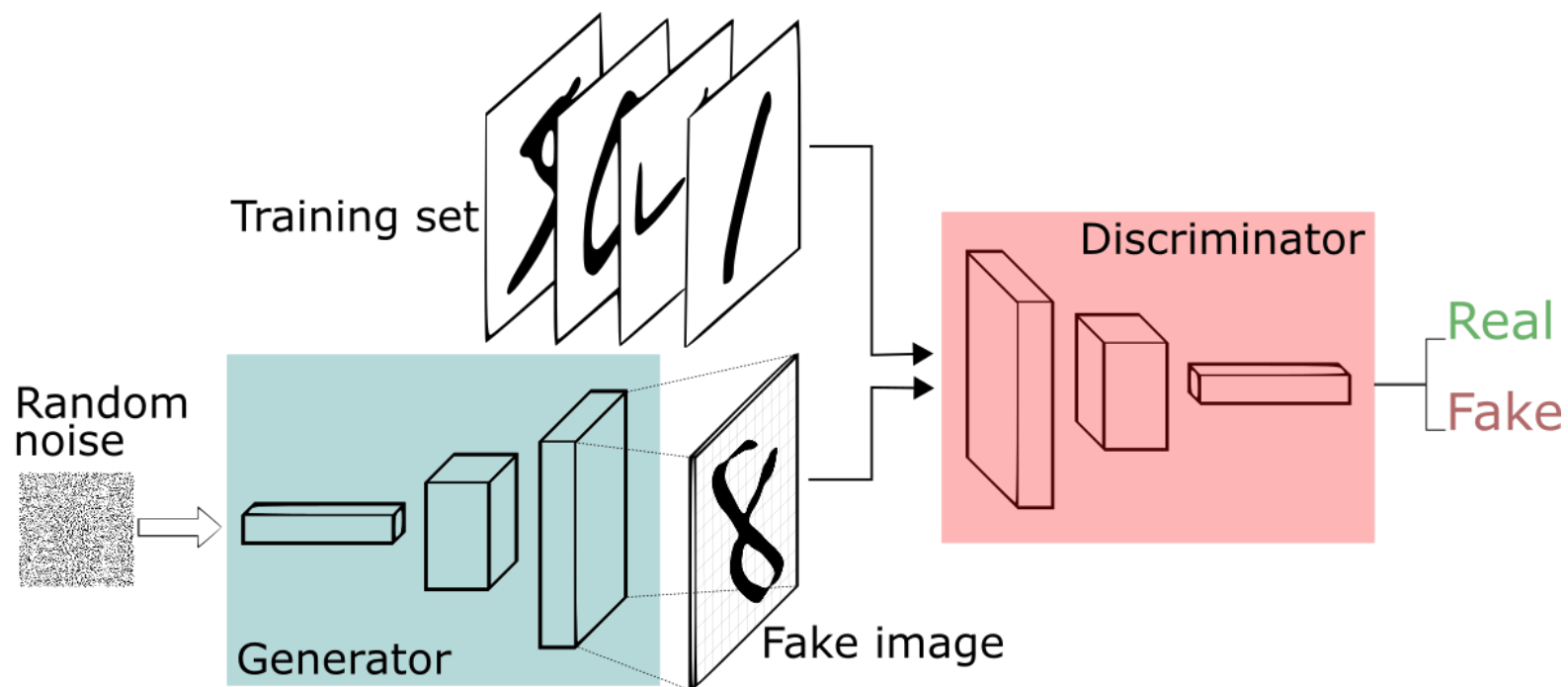
Output Sequence

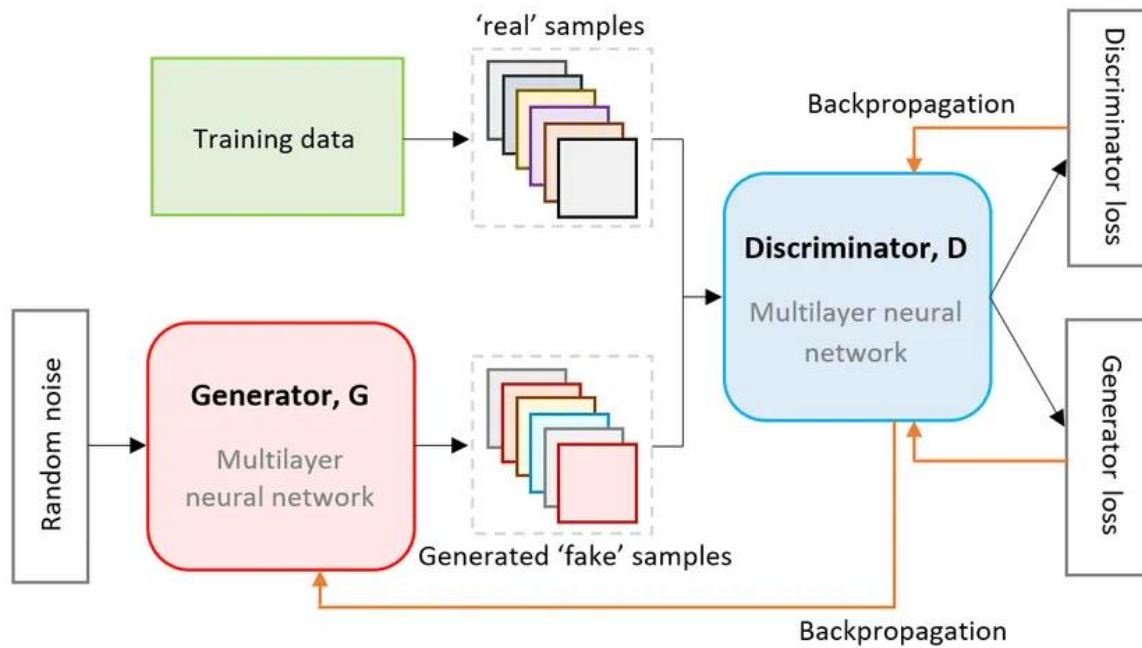
Y

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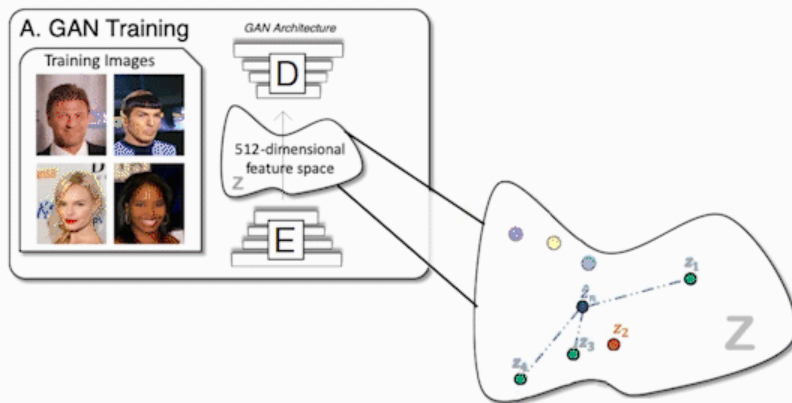
gan



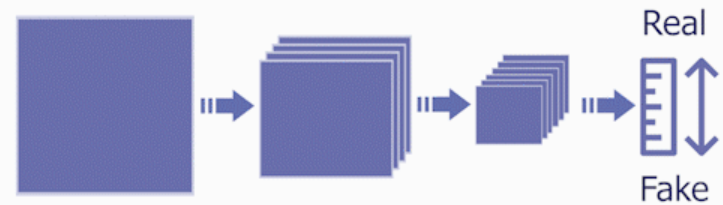


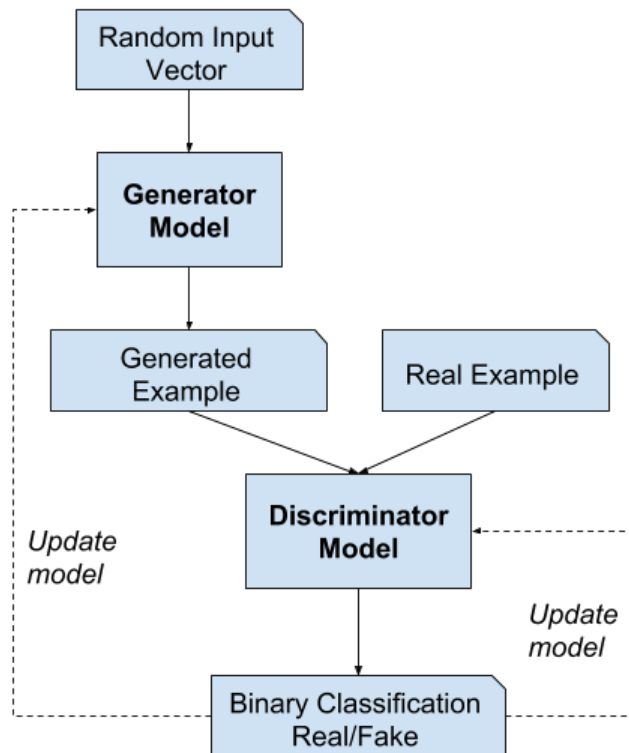
@whats_ai

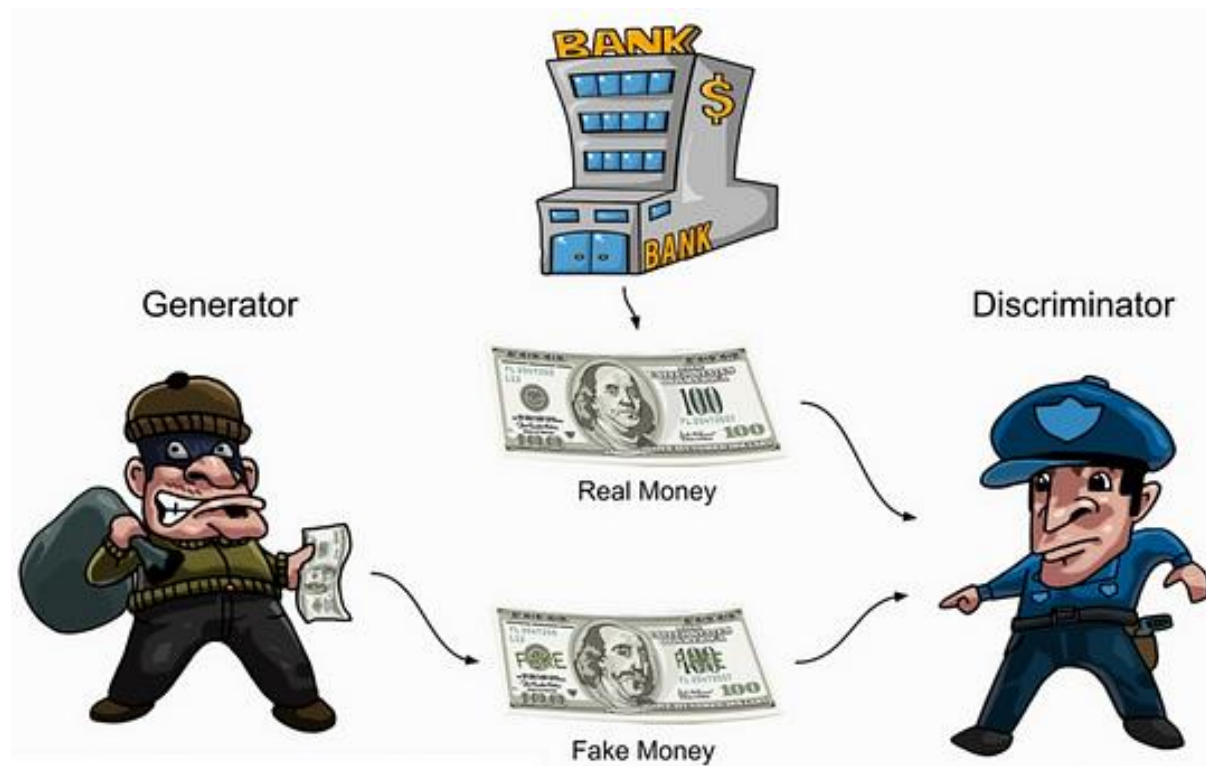
Generator



Discriminator







Self-Attention

Self-Attention is a mechanism used in neural networks, particularly in the Transformer model, to capture the importance of different words in a sentence or sequence. The mathematical formula for Self-Attention is as follows:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Where:

- Q is the Query matrix
- K is the Key matrix
- V is the Value matrix
- d_k is the dimension of K (or Q)

Note:

the Matrcies are learnble wieghts

What is a language model

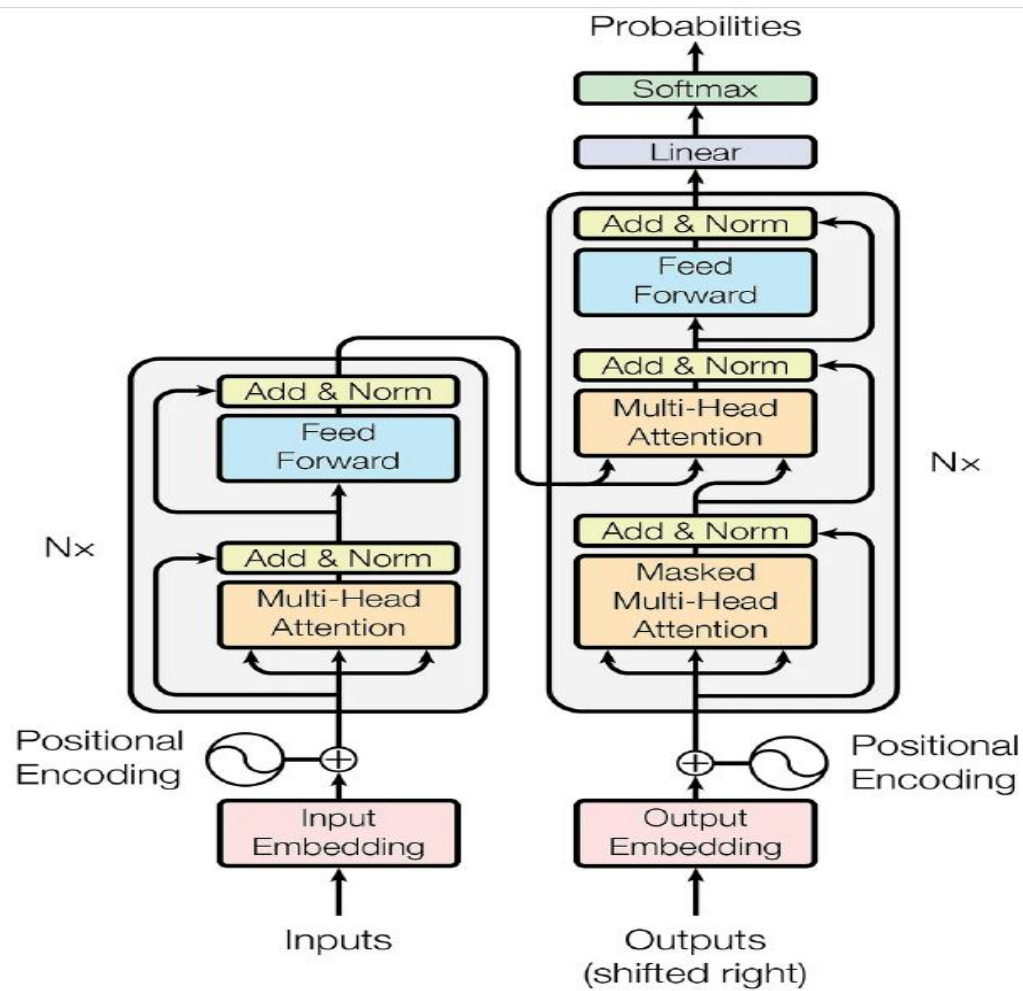
- The classic definition of a language model (LM) is a **probability distribution over sequences of tokens**.
- Suppose we have a **vocabulary** V of a set of tokens.
- A language model p assigns each sequence of tokens $x_1, \dots, x_L \in V$
- a probability (a number between 0 and 1) .
- $p(x_1, \dots, x_L)$.

The probability intuitively tells us how “good” a sequence of tokens is. For example, if the vocabulary is $V=\{\text{ate},\text{ball},\text{cheese},\text{mouse},\text{the}\}$, the language model might assign (demo):

$$p(\text{the},\text{mouse},\text{ate},\text{the},\text{cheese})=0.02,$$

$$p(\text{the},\text{cheese},\text{ate},\text{the},\text{mouse})=0.01,$$

$$p(\text{mouse},\text{the},\text{the},\text{cheese},\text{ate})=0.0001.$$



"This is a input text."

Tokenization

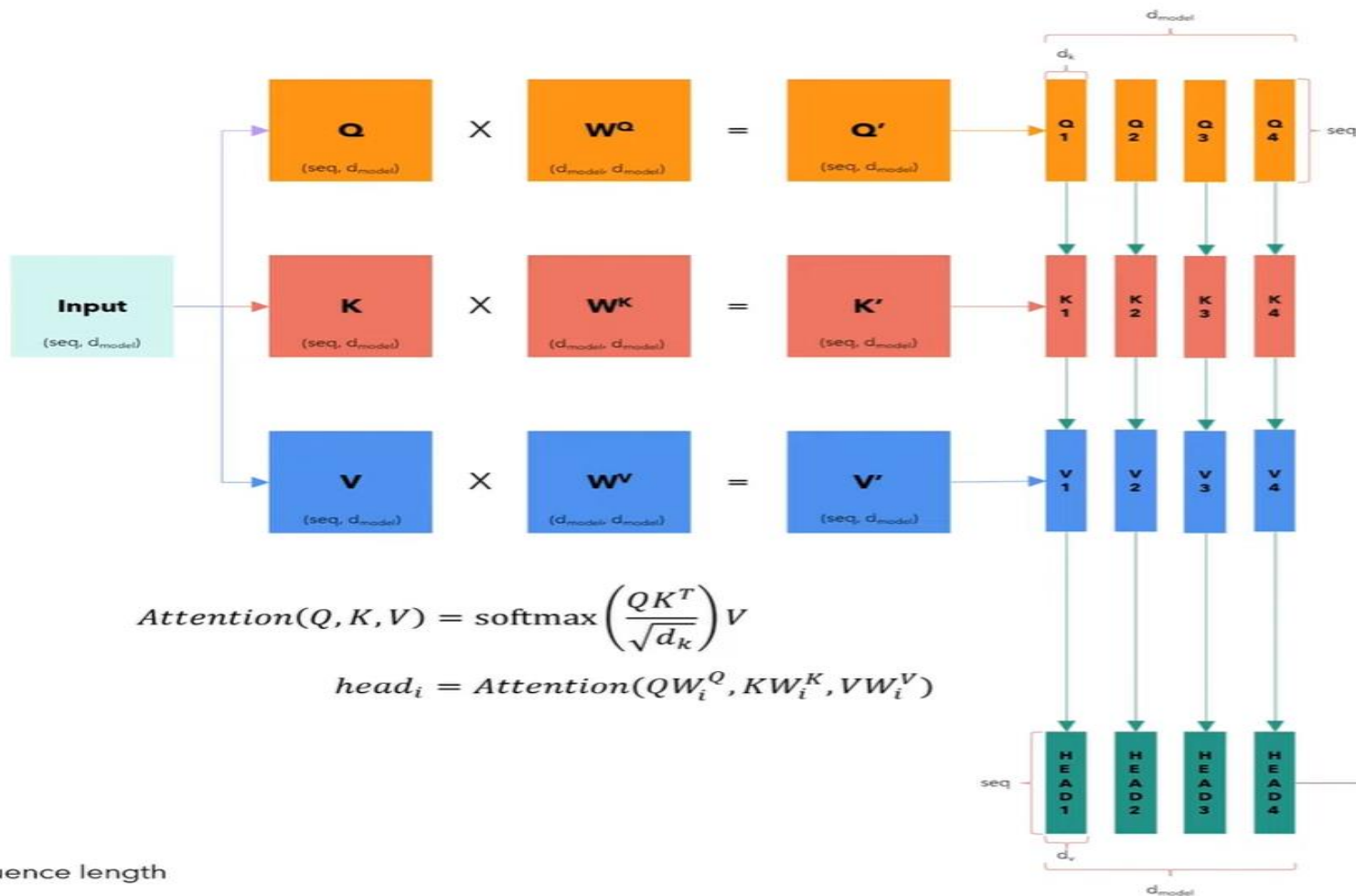


[CLS]	This	is	a	input	.	[SEP]
101	2023	2003	1037	7953	1012	102

Embeddings



0.0390,	-0.0558,	-0.0440,	0.0119,	0069,	0.0199,	-0.0788,
-0.0123,	0.0151,	-0.0236,	-0.0037,	0.0057,	-0.0095,	0.0202,
-0.0208,	0.0031,	-0.0283,	-0.0402,	-0.0016,	-0.0099,	-0.0352,
...	...	...	...	...	...	...



$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

$$head_i = Attention(QW_i^Q, KW_i^K, VW_i^V)$$

$$MultiHead(Q, K, V) =$$

sequence length

size of the embedding vector

number of heads

d_{model} / h

Visual view / word embedding, self attention, Transformer Model

- <https://ig.ft.com/generative-ai/>
- <https://poloclub.github.io/transformer-explainer/>
- <https://bbycroft.net/llm>
- <https://jalammar.github.io/illustrated-transformer/>

- <https://huggingface.co/models>

